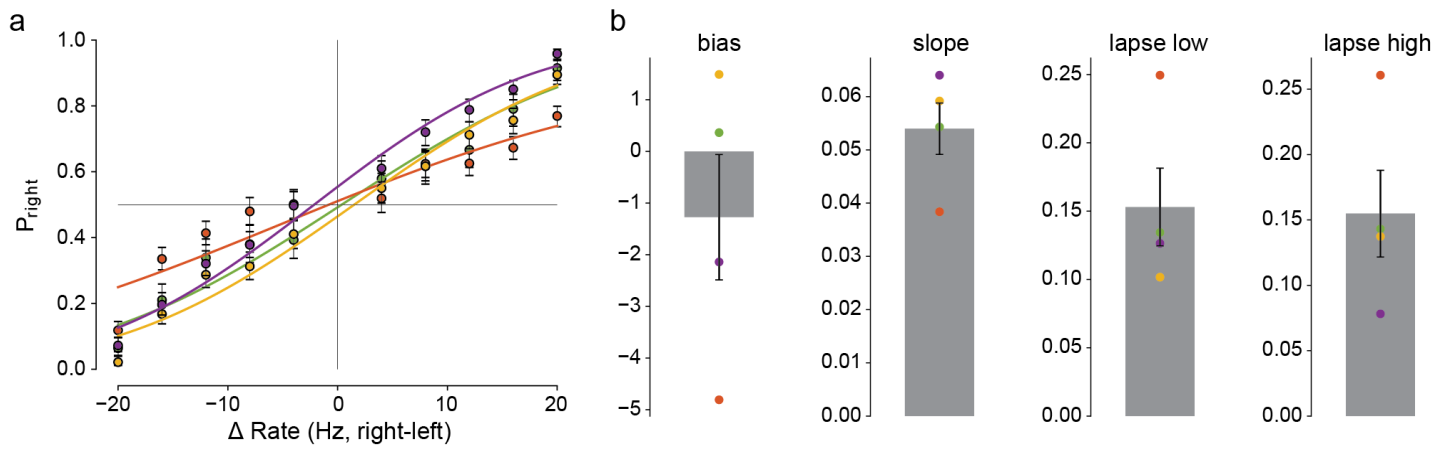


Neuron, Volume 113

Supplemental information

Spontaneous movements and their relationship to neural activity fluctuate with latent engagement states

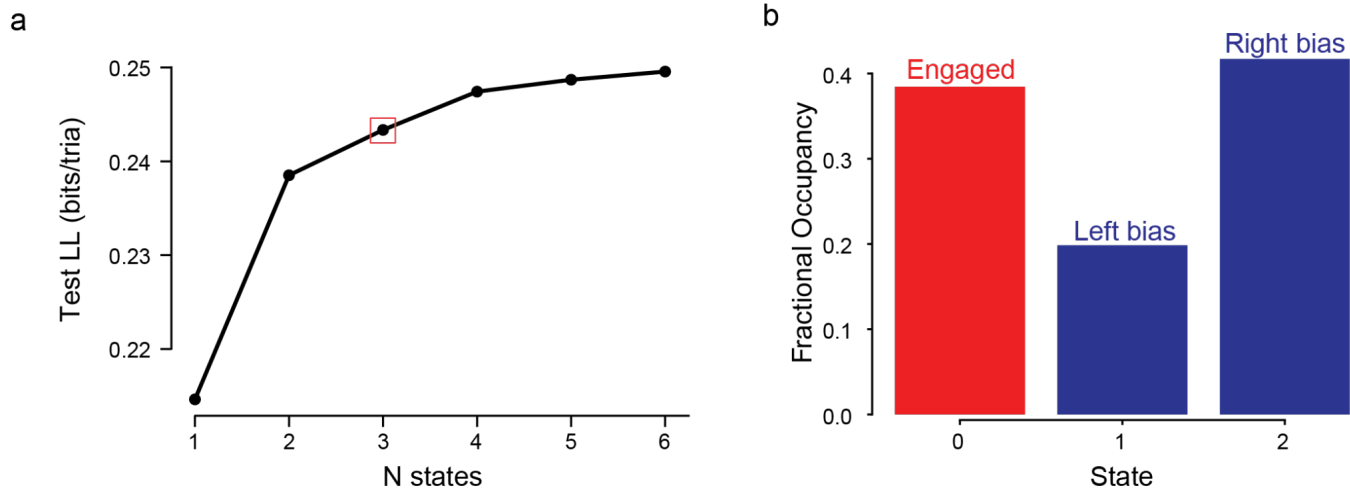
Chaoqun Yin (尹超群), Maxwell D. Melin, Gabriel Rojas-Bowe, Xiaonan Richard Sun, João Couto, Steven Gluf, Alex Kostiuk, Simon Musall, and Anne K. Churchland



Supplementary figure 1: Psychometric function parameters. Related to figure 1.

a) Behavioral data and psychometric functions for individual mice; color indicates mouse identity. points denote the probability of rightward choices; error bars indicate Wilson 95% confidence intervals.

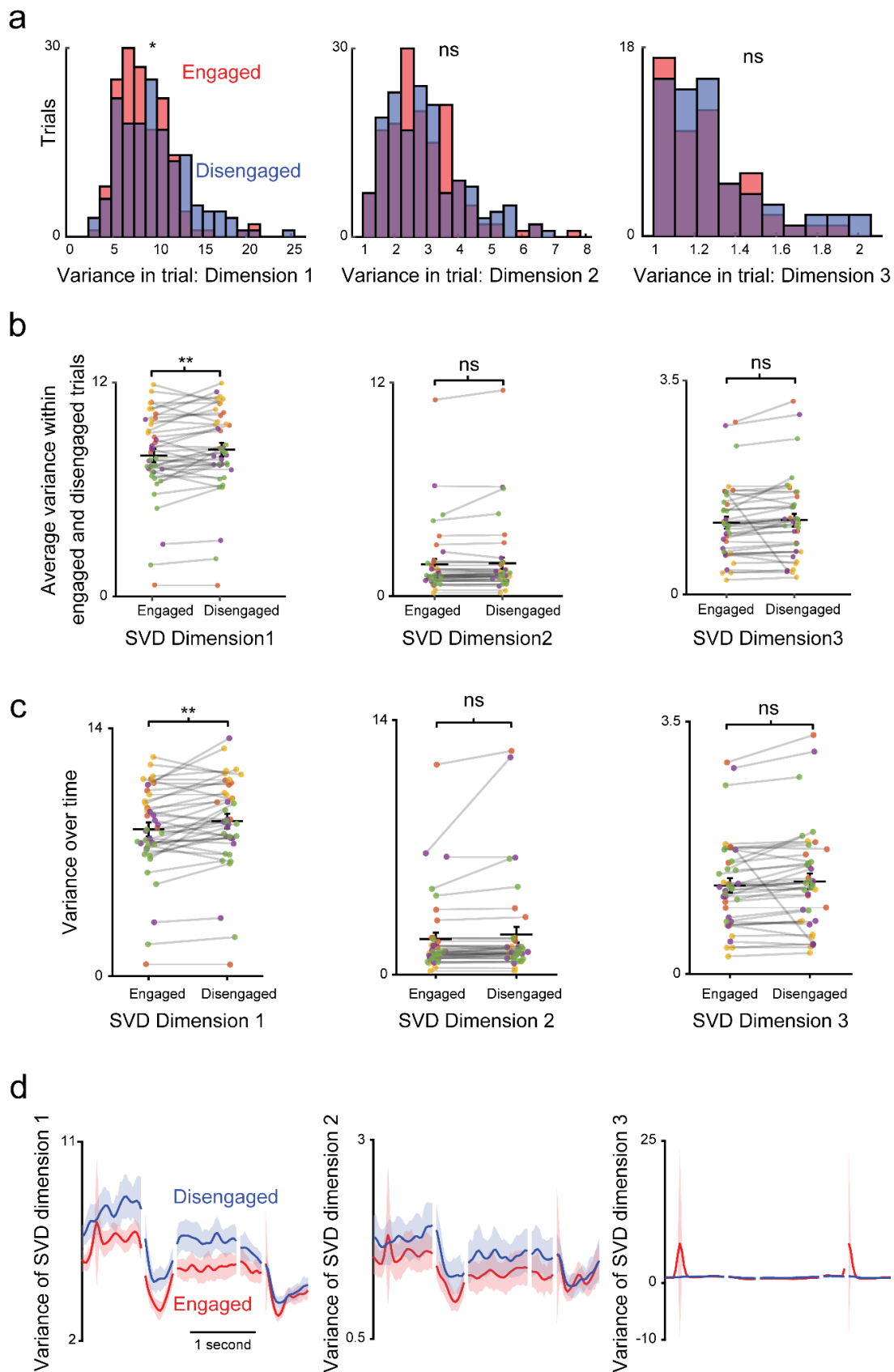
b) Psychometric function parameters extracted from the functions shown in (a). Bars denote mean $\pm$ SEM.



Supplementary figure 2: GLM-HMM cross-validation and fractional state occupancy. Related to figure 1.

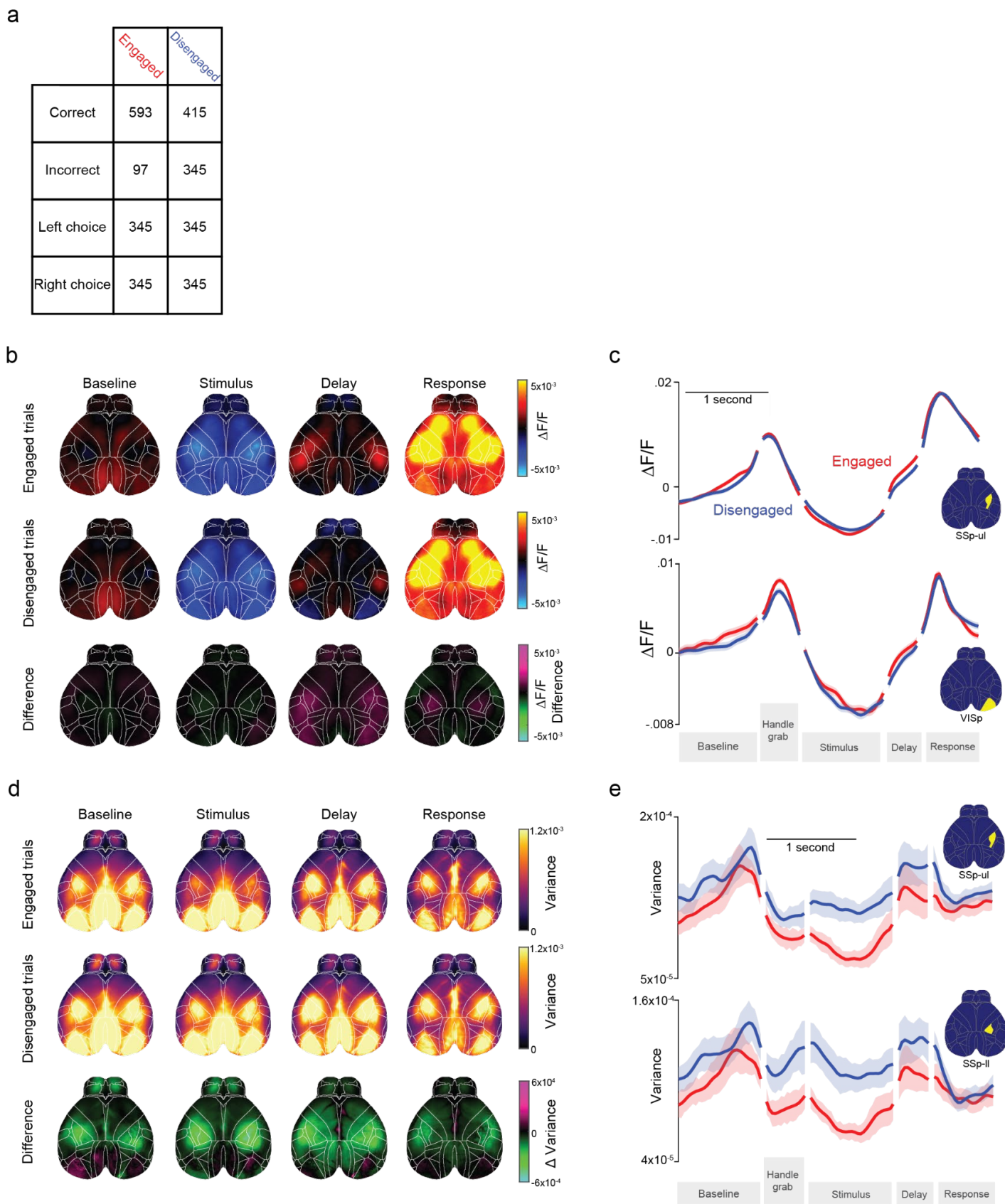
a) 10-fold cross validated log-likelihood (bits/trial) for increasing numbers of latent states. Grid search was performed over three parameters (n_{states} , α , and σ). The best performing model for each value of n_{states} is shown. Red square indicates the number of states used for subsequent analysis.

b) Fractional occupancy for a 3-state model.



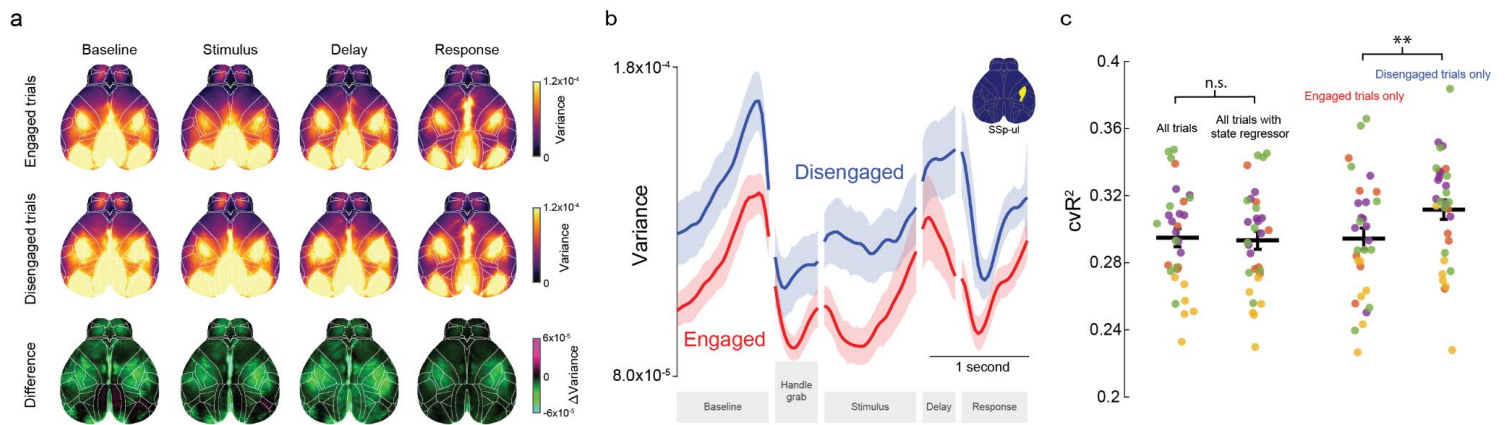
Supplementary figure 3: Disengaged trials show higher variance in the first SVD dimension of neural activity. Related to figure 2.

- a)** Histograms for an example session denoting the variance in widefield SVD components across time, within individual engaged and disengaged trials.
- b)** The average variance of the first three SVD dimensions of widefield neural activity is plotted for both engaged and disengaged trials from an example session. Trials counts were matched within sessions (paired data points from the same session are denoted by black lines). Variance in the first SVD dimension is greater in disengaged trials than engaged trials ($p < .01$, linear mixed effects model with mouse as a random effect). Variance in subsequent dimensions is not significantly different. Color shades denote individual mice.
- c)** Same as **b**, but data points now indicate total variance across all engaged or disengaged timepoints, rather than the average variance within engaged or disengaged trials.
- d)** Across-trial variance for the top 3 SVD dimensions. Mean across-trial variance $\pm$ SEM are shown.



Supplementary figure 4: State-dependent differences in cross-trial variance are not due to state-dependent choice differences. Related to figure 2.

- a)** Trial numbers for b-e. Left and right choice trials are equalized to ensure state-dependent differences are not driven by state-dependent choice side preferences.
- b)** Heatmaps of average activity during each trial epoch, averaged over 19 sessions from 4 mice. Top row: engaged trials. Middle row: disengaged trials. Bottom row: difference (engaged activity minus disengaged activity). Purple indicates more activity during the engaged state, green indicates more activity during the disengaged state.
- c)** Average activity over time for two example regions. Error: $\pm$ SEM across sessions.
- d)** Heatmaps of average variance across trials. Higher values denote more trial-to-trial differences in neural activity. Row conventions are the same as in (e).
- e)** Across-trial-variance plotted for two example regions. Error: $\pm$ SEM across sessions.

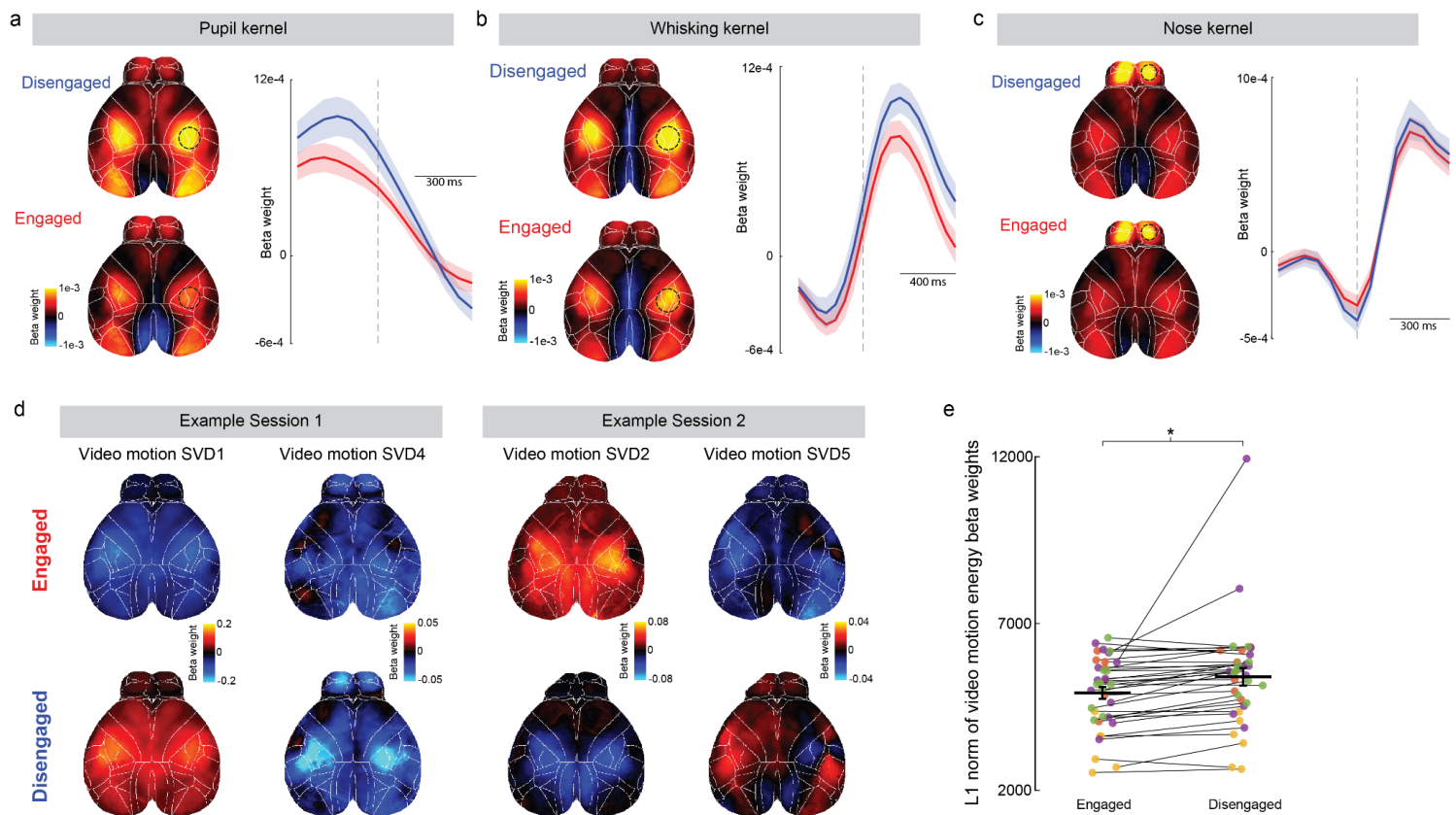


Supplementary figure 5: State-dependent differences in cross-trial variance are not due to pooling left- and right-biased trials. Related to figure 2.

a) Heatmaps of average variance across trials, taken from only left choice trials. Higher values denote more trial-to-trial differences in neural activity.

b) Across-trial-variance plotted for one example region, again for only left choice trials. Error: $\pm$ SEM across sessions.

c) Cross-validated variance explained plotted for multiple sessions. The same convention as Fig. 3d, except right- and left-biased states were not grouped within sessions. We instead used only one biased state (the one with more trials for a given session). Left two columns, total variance explained by linear encoding models trained on all trials, with and without a state regressor. Error bars denote mean $\pm$ SEM. Scatter points denote individual sessions, with marker colors denoting mice. Linear mixed-effects model with mouse identity as a random effect, $p = 0.79$. Right two columns, linear encoding models trained on engaged or disengaged trials from matched sessions. Linear mixed-effects model with mouse identity as a random effect, $p < .01$.



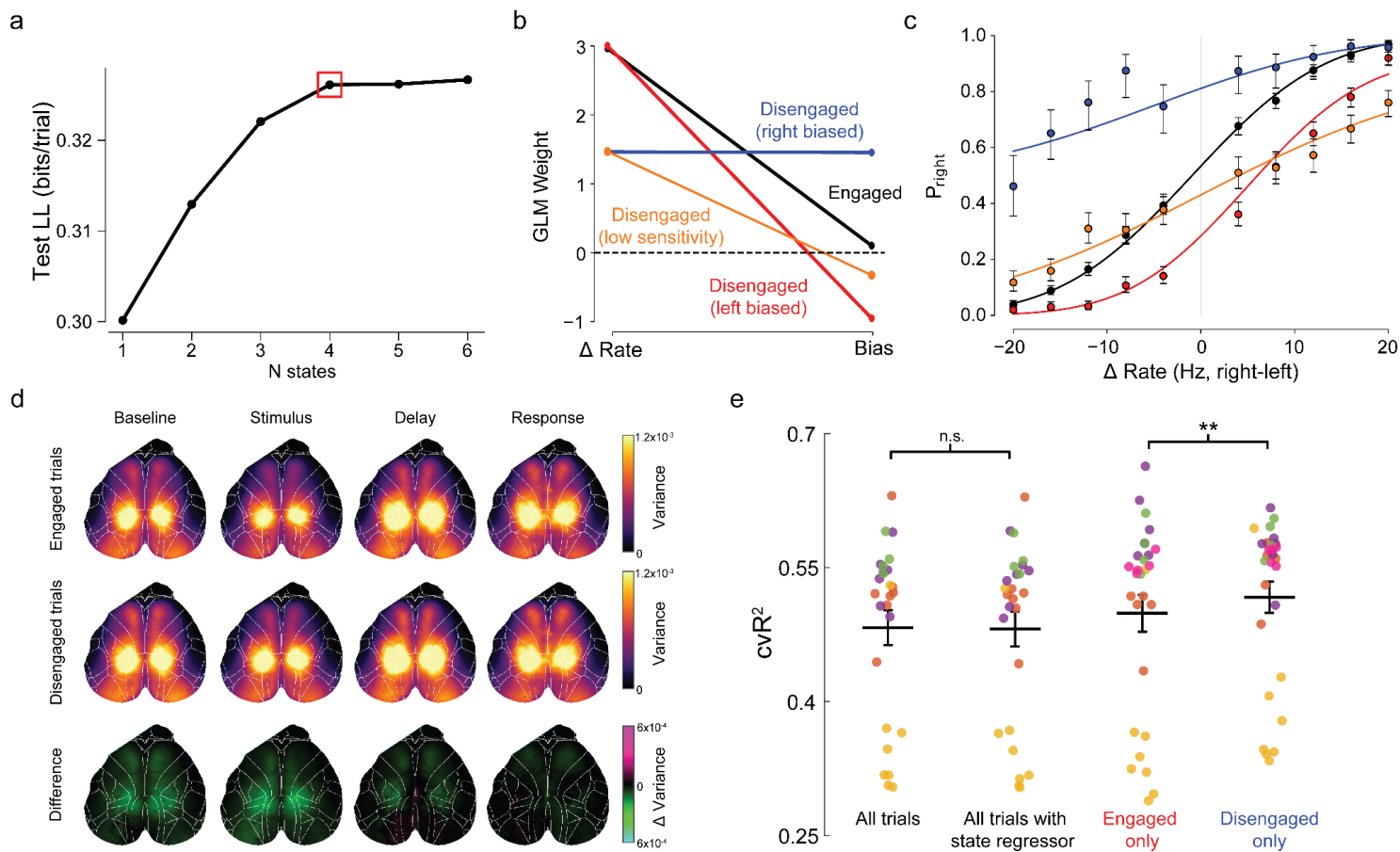
Supplementary figure 6: Movement kernels change modestly with behavioral state. Related to figure 2 & 3.

a) Left, heatmap of beta weights for the pupil kernel in engaged and disengaged states. Blue and red lines denote the time series kernel from the region of interest denoted in **a** with a back circle. Error bars denote $\pm$ SEM. Grey dashed line indicates the time of event occurrence. $p < 0.001$, linear mixed effects model with mouse identity as random effect. Kernel norm p -value $< .001$.

b-c) Same as in **a** but for the whisking, and nose kernels respectively. Kernel norm p -values = 0.61 and 0.17 respectively.

d) Kernel heatmaps for video motion energy SVD components. Headmaps are shown for individual sessions, since behavior video SVD spatial components vary across sessions.

e) L1-norm of all video motion energy SVD beta weights. Each dot denotes the norm for a particular session, with color indicating the mouse identity.



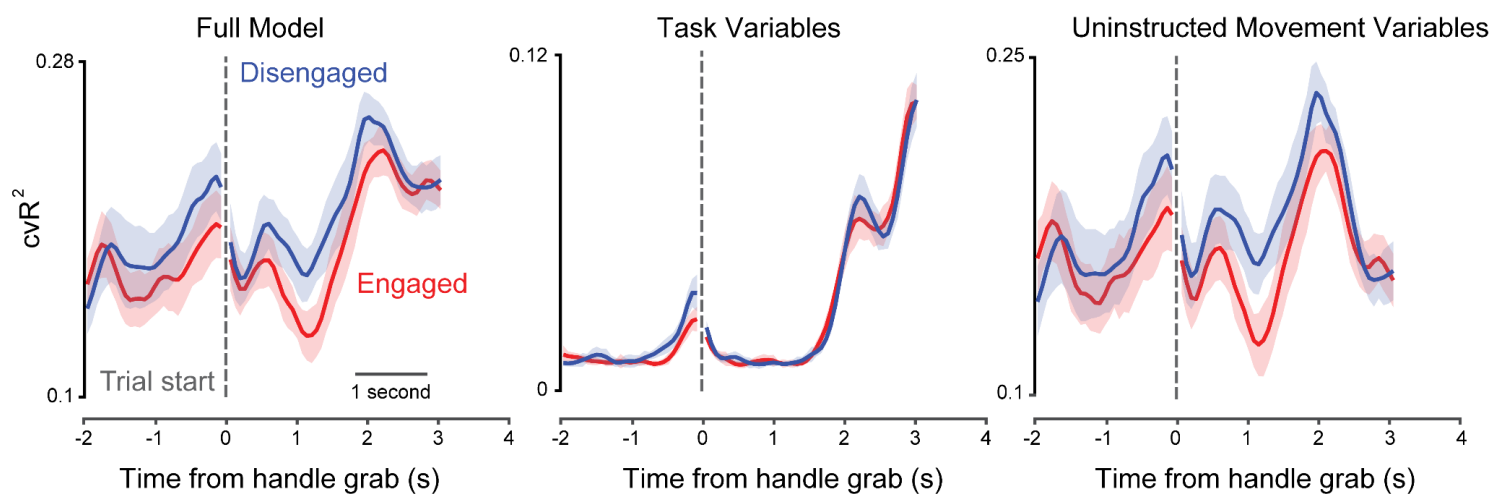
Supplementary figure 7: In PT-labeled animals, disengagement is also associated with more neural variance and a higher percentage of neural variance that can be explained by an encoding model. Related to figure 3.

a) A 4-state HMM-GLM is fitted to the behavior data of 5 PT-labeled mice.

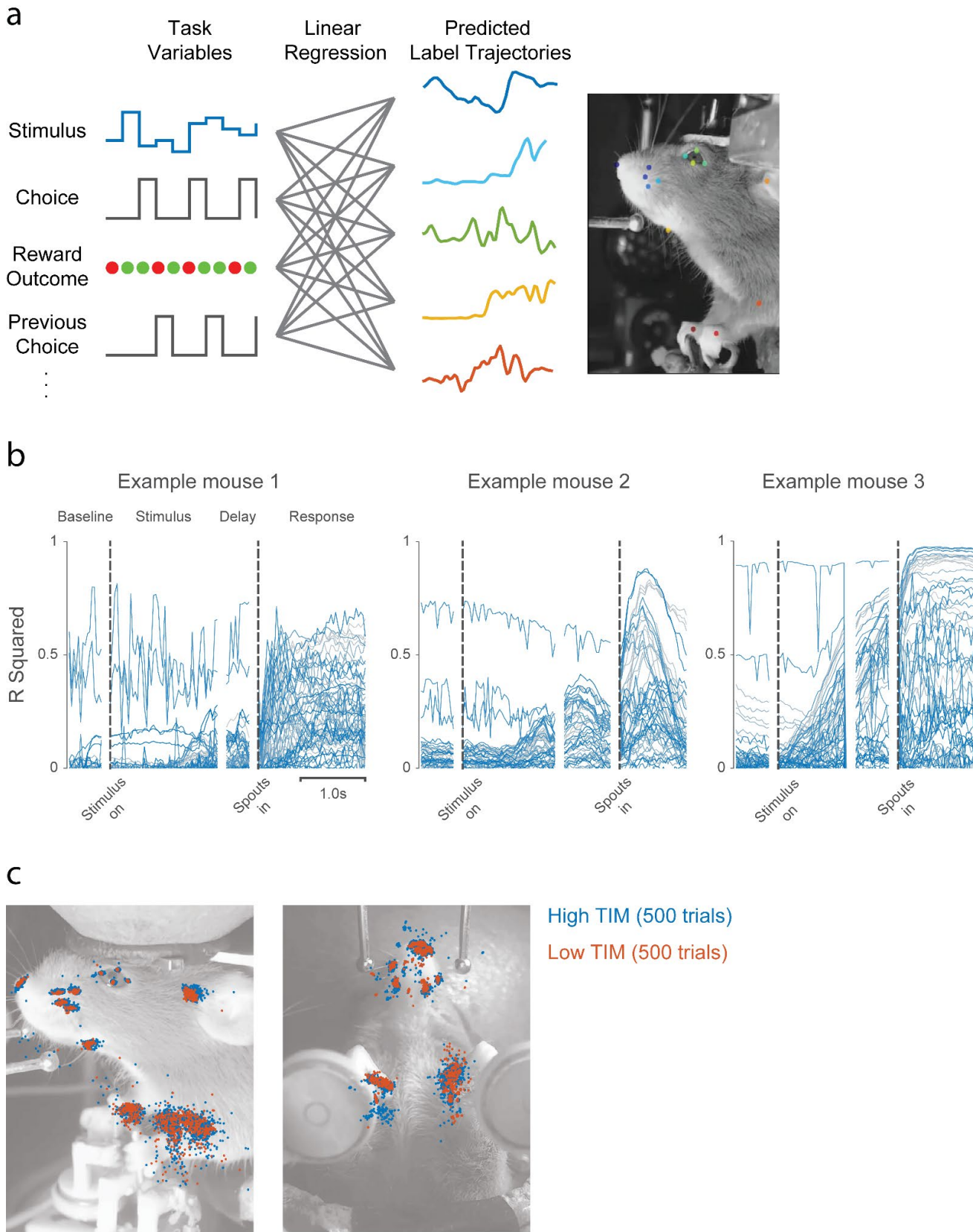
b-c) The model finds one engaged state, two disengaged states with left-/right-choice biases, and a 4th state with lower sensitivity. We grouped 3 suboptimal states and termed these “disengaged”.

d) Neural variance is higher in PT neurons during disengagement.

e) Left two columns: a single extra state regressor does not account for more variance (as observed in the left column of Fig. 3d). Right two columns: the encoding model captures a higher percentage of neural variance (indicated by higher cvR^2) during disengagement ($p = 0.0067$), mixed-effect model was fitted to the data with mouse identity as the random effect.). This increase in explained variance during disengagement is similar to what we reported in Fig. 3d (right columns).



Supplementary figure 8 (related to figure 3 & 4): The increase in the cvR^2 of the full model and the uninstructed movement model during disengagement (Fig. 3c,e) is replicated after replacing the video data in our linear encoding model with DeepLabCut tracking data.



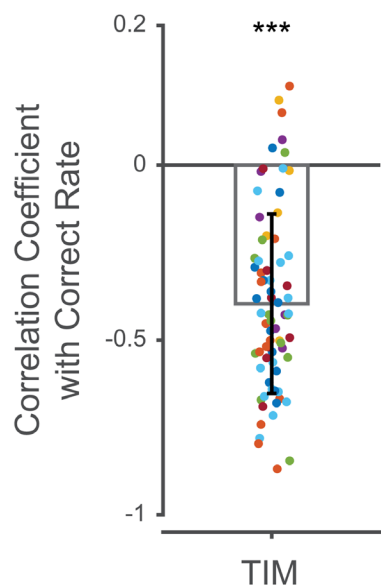
Supplementary figure 9: Linear regression is used to predict the movement trajectory of each body part. Related to figure 5.

a) Task variables (the left column) are used as regressors to predict the position of each DeepLabCut label at each frame (the right column, each curve denotes the predicted trajectory of one DeepLabCut label). Each body part was

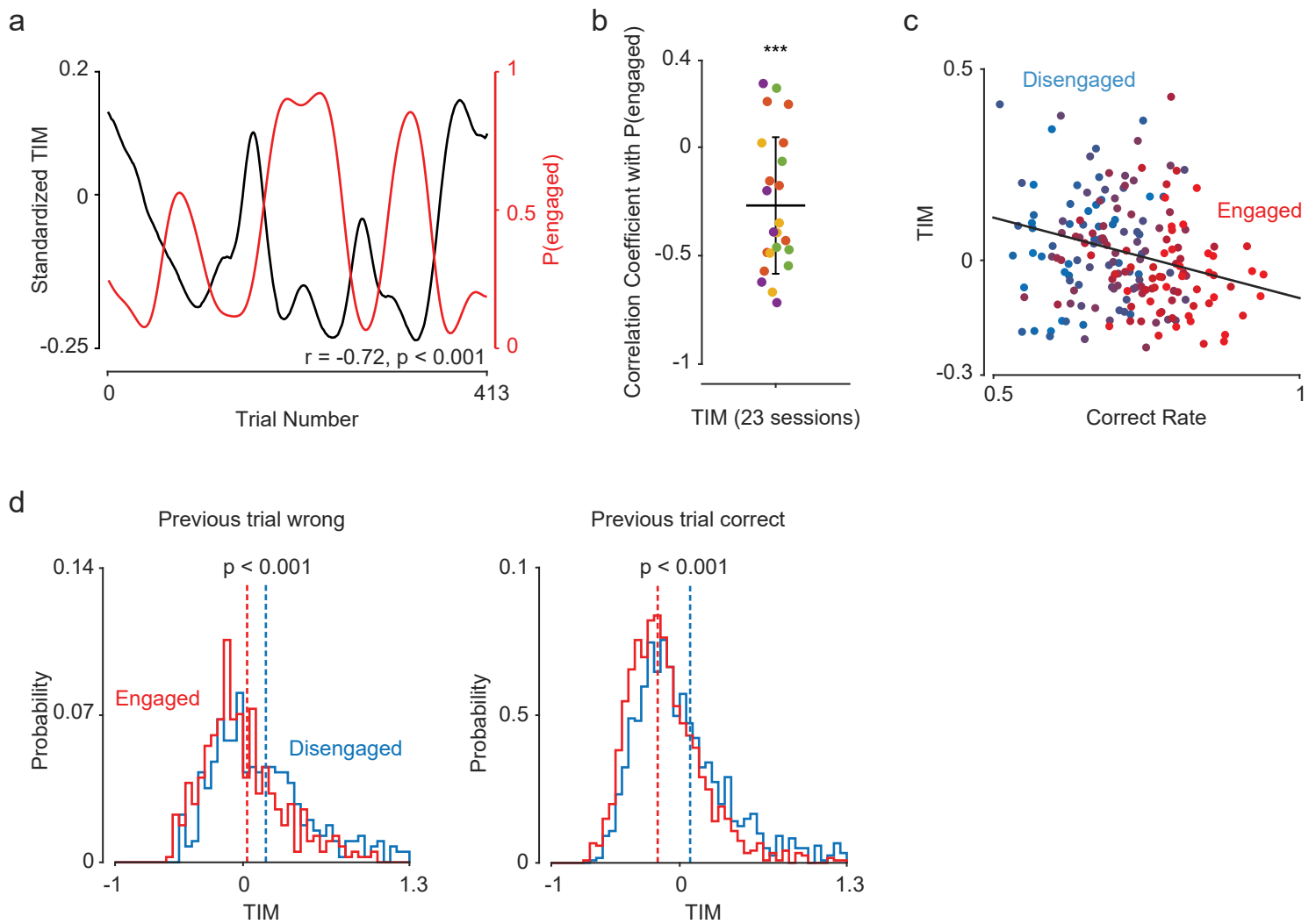
fitted with a separate model at each time point. TIM is defined as the distance between the predicted and actual positions (Fig. 5a).

b) 3 panels show the R^2 values across time of all DeepLabCut labels (x and y are plotted separately) of 3 example animals. Cross-validated linear model captures significant variance in DeepLabCut-tracked movement.

c) The DeepLabCut label positions of 500 trials with high TIM and 500 trials with low TIM at one example frame (in the stimulus epoch) from one example animal are plotted. The trials with high TIM values distribute in a bigger area.



Supplementary figure 10 (related to figure 5): TIM-task performance correlation of 9 additional animals (73 sessions). These animals were used in optogenetic inactivation experiments (inactivation trials would interrupt continuous HMM state) or had cells labeled other than Emx-expression neurons. Thus, these animals were not included in the GLM-HMM analysis. All optogenetic inactivation trials were removed before calculating TIM. As Fig. 5d, the correlation coefficients of sessions are plotted with dots. The colors denote different animals. The bars indicate the averaged coefficients across sessions. We again observed negative correlation between TIM and correct rate (Correlation coefficients of 73 sessions = -0.3962 (mean) ± 0.2566 (standard deviation), p of mixed-effects model = $4.7796e-21$).



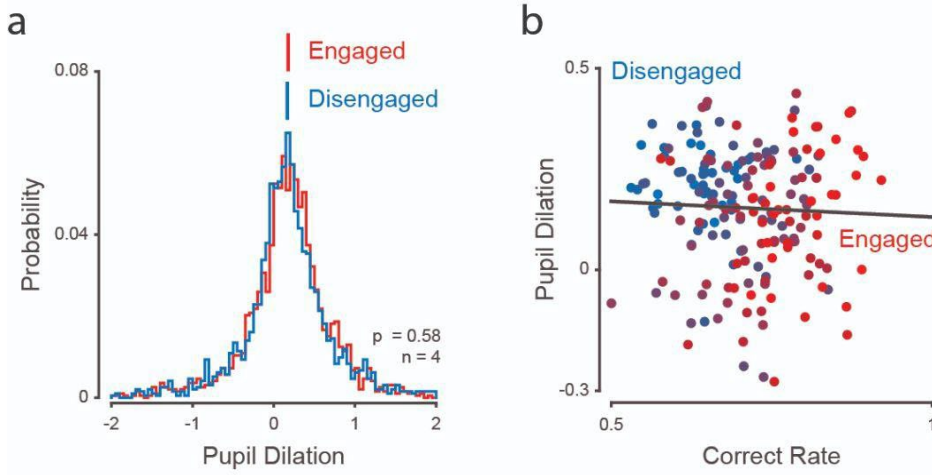
Supplementary figure 11: TIM remains a negative indicator of engagement after removing outcome-related regressors from TIM calculation or controlling for the outcomes of previous trials. Related to figure 5.

a) The fluctuation of smoothed TIM and $P(\text{engaged})$ in an example session. In this case, TIM was calculated without outcome-related regressors (same session as in Fig. 5c; $r = -0.717$, $p = 2.114\text{e-}66$).

b) The negative TIM-engagement correlation persists after removing outcome-related regressors. Correlation coefficients of 23 sessions = -0.2687 (mean) ± 0.3149 (standard deviation). A linear mixed-effects model was fitted to the correlation coefficients with mouse identity as a random effect. The correlation coefficients are significantly below zero ($p = 3.8575\text{e-}04$). Both TIM and $P(\text{engaged})$ were smoothed with a 50-trial long Gaussian filter before testing the correlation coefficients.

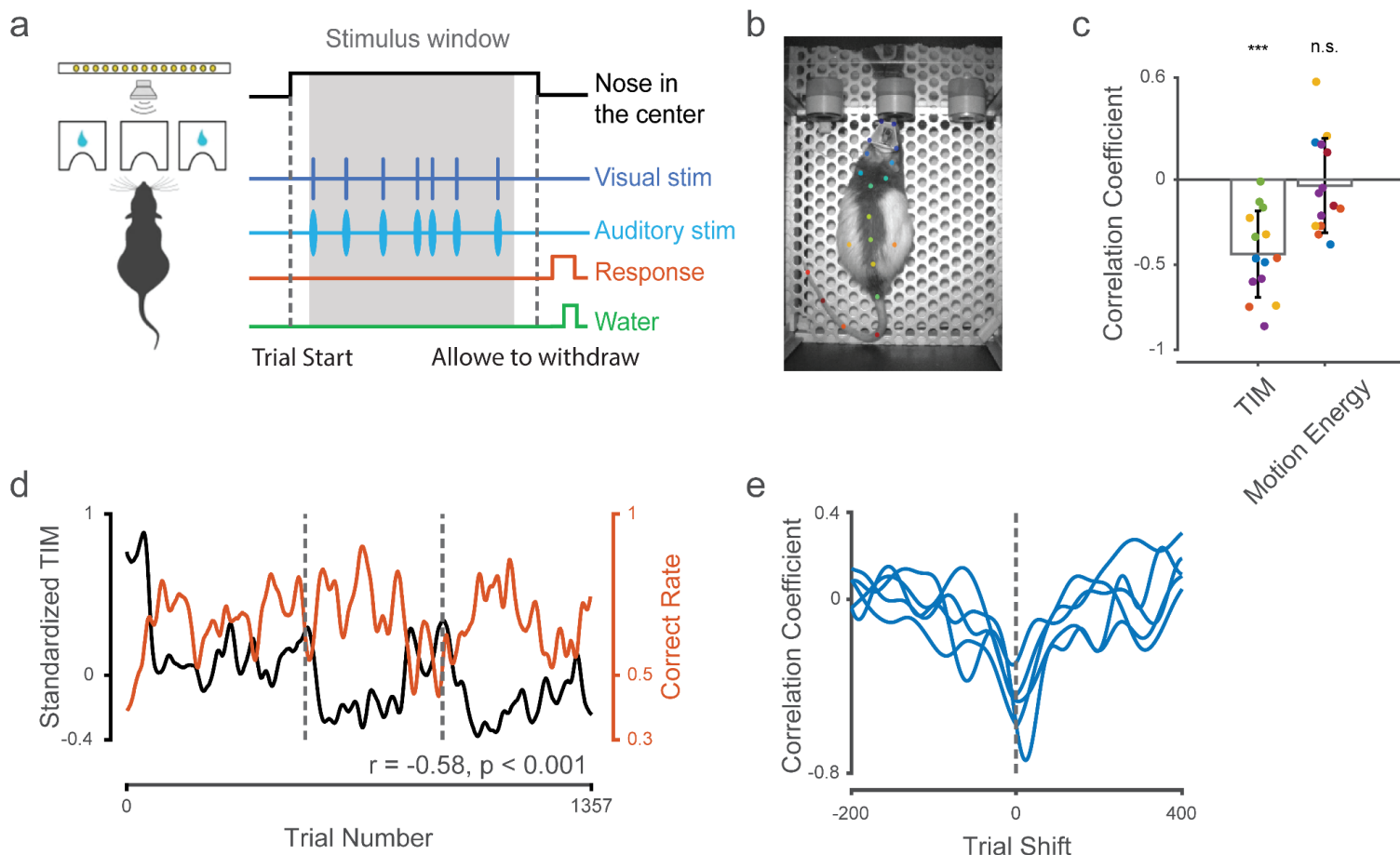
c) The negative TIM-engagement and TIM-performance correlations persist after removing outcome-related regressors. (TIM-correct rate correlation: $r = -0.2520$, $p = 0.0004$. The data from 4 animals was used.)

d) Trials were grouped based on the previous trial's outcome. Dashed lines denote the mean TIM for engaged and disengaged trials. After controlling the outcome of the previous trial, TIM is still higher during disengagement (red dashed line to the right of blue dashed line). (Left: TIM of engaged trials = 0.0317 (mean) ± 0.3965 (standard deviation), TIM of disengaged trials = 0.1783 ± 0.4446 , unpaired t-test, $p = 1.1484\text{e-}06$, 794 trials from 4 animals are included; Right: TIM of engaged trials = -0.1108 ± 0.3112 , TIM of disengaged trials = 0.0382 ± 0.4219 , p of unpaired t-test = $1.5435\text{e-}22$, 2412 trials from 4 animals are included.)



Supplementary figure 12: Pupil dilation is not correlated with P(engaged) or task performance. Related to figure 6.

- a)** Distribution of pupil dilation of engaged/disengaged trials. The red/blue lines denote the mean pupil dilation of engaged/disengaged trials respectively. No smoothing across trials was applied. There is no significant difference in pupil dilation (the pupil diameter difference between the stimulus and baseline epochs) between engaged and disengaged states (pupil dilation of engaged trials = 0.1783 (mean) $\pm$ 0.5670 (standard deviation), pupil dilation of disengaged trials = 0.1676 ± 0.6217 , p of unpaired t-test = 0.5755 . The data from 4 animals was used.).
- b)** Pupil dilation and correct rate were both smoothed with a 50-trial long Gaussian filter and downsampled every 50 trials. Each dot denotes a trial. The color of each dot was assigned based on the value of P(engaged). Pupil dilation does not correlate with task performance (Pupil dilation-correct rate correlation: $r = -0.0455$, $p = 0.5318$. The data from 4 animals was used.).



Supplementary figure 13: TIM is a negative indicator of performance in freely-moving rats. Related to figure 5 & 7.

a) Freely-moving rats were trained to do a multisensory perceptual decision-making task. Animals were trained to report if the number of stimulus events (clicks, flashes, or both) during the stimulus window was higher or lower than a learned category boundary (12 events). Choices to the right (left) port were rewarded when the stimulus was higher (lower) than 12 events.

b) 19 body parts were tracked with DeepLabCut. Color dots denote DeepLabCut labels.

c) The correlation between TIM/motion energy and task performance. Dots represent 14 individual sessions, with colors indicating the 5 animals from which the sessions were collected. Bars show the averaged coefficients across sessions. Both TIM and correct rate were smoothed with a 30-trial long Gaussian filter before calculating the correlation coefficients. We found a negative correlation between TIM and task performance, though the motion energy was independent of task performance. (Left: TIM-correct rate correlation = -0.4381 (mean) ± 0.2539 (standard deviation). A linear mixed-effects model was fitted to the correlation coefficients with rat identity as a random effect. $p = 1.244e-04$. Right: Motion energy-correct rate correlation = -0.0355 ± 0.2773 . Mixed-effects model $p = 0.6275$.)

d) The fluctuations of TIM and task performance in 3 example sessions from one rat. Both TIM and correct rate were smoothed with a 30-trial long Gaussian filter. ($r = -0.5811, p = 2.346e-123$)

e) Trial shift test shows that there is no time lag in the negative TIM-performance correlation. Each line indicates, for a single animal, the correlation coefficient for the time lag indicated on the horizontal axis.

Instructed Movements	Uninstructed Movements	Task Variables
Left handle grab	Piezo body sensor (analog)	Time
Right handle grab	Piezo body sensor (digital)	Choice
Left spout lick	Whisking (analog)	Reward
Right spout lick	Whisking (digital)	Previous choice
	Nose movements (analog)	Previous reward
	Nose movements (digital)	Next choice
	Pupil (fast analog)	Water delivery
	Pupil (fast digital)	Handle movement sound
	Pupil (slow analog)	Left click
	Facial motion (analog)	Right click
	Facial motion (digital)	Left stimulus
	Video SVD (orthogonalized to spout and handle movement)	Right stimulus
	Video motion energy SVD (orthogonalized to spout and handle movement)	

Supplementary table 1: Encoding model regressor groups. Related to figure 3.